ETSN11, Final Exam 2024-11-01, Solutions

This document includes our proposed solutions. The solutions are quite short and may not have given full points on the exam. Other solutions, which we have not thought about, can also be correct.

General advice: In **Part A,** questions 6-8, you need to correctly describe the specific terms and concepts to get full points. In **Part B**, you need to motivate your chosen formulas and calculations to get full points. In **Part C**, you need to give a suitable solution to the specific scenario. To get full points, this solution must both address the requirements and challenges in the scenario **and** include specific protocols and technologies.

Part A: Technologies and architectures (30 points)

In the **multiple-choice questions**, you get points for each correct answer, but minus 1 point for each incorrect answer. You cannot get a lower score than 0 in the multiple-choice questions.

1. Which of the following statements are true for **Bluetooth Low Energy (BLE)**? (3p)

a. BLE is a network technology aimed at non-critical applications.
b. BLE is a network technology aimed at time-critical applications.
c. Animal and environmental monitoring are some targeted applications for BLE.
d. Smart homes and industrial sensors and actuators are some targeted applications BLE.
e. Self-driving vehicles, robots and drones are some targeted applications for BLE.
f. BLE is based on a decentralized network architecture.
g. BLE is based on a centralized network architecture.
h. BLE uses delay-tolerant networking.
i. BLE allows for multi-hop data dissemination.
j. The data dissemination in BLE can only be performed in single-hop.

Clarifications: The BLE standard uses a mesh architecture, that is, it is based on a decentralized architecture with multi-hop communication. BLE is developed for non-critical applications. However, due to the limited communication range that Bluetooth has, BLE is not targeting outdoor sensor applications as animal or environmental monitoring.

2. Which of the following statements are true for **Centralized network architectures**? (3p)

a. ZigBee is based on a centralized network architecture.
b. 5G is based on a centralized network architecture.
c. Starlink is based on a centralized network architecture.
d. LoRa is based on a centralized network architecture.
e. WiFi Direct is based on a centralized network architecture.
f. In a centralized network architecture, nodes can communicate with a base station / access point using single-hop communication.
g. In a centralized network architecture, nodes can communicate with a base station / access point using multi-hop communication.
h. "Centralized" means that nodes only communicate with one base station / access point.
i. "Centralized" means that nodes are geographically close to each other.

j. “Centralized” means that the base station / access point controls the medium access for nodes.
k. “Centralized" means that nodes communicate via a base station / access point, not directly with each other.

Clarifications: In a centralized network architecture, nodes only communicate in single-hop with a base station/access point, not directly with each other (then it is an ad-hoc network). 5G and LoRa both are based on a centralized network architecture. ZigBee uses a mesh architecture, which is a decentralized network architecture. Wifi-direct allows for ad-hoc connections, that is, a decentralized network architecture. Starlink uses a centralized network architecture from the users’ perspective (The satellites may be organized as a mesh network, but very little is known about this, so I have skipped this in the course lectures).

3. Which of the following statements are true for **sensor networks**? (3p)

a. There is always only one sink in the network.
b. There are always multiple sinks in the network.
c. Each sensor has a MAC/physical address.
d. Each sensor has an IP-address.
e. The aim of a sensor network is to transport data to a sink.
f. The aim of a sensor network is to communicate with a back-end system.
g. The aim of a sensor network is to allow nodes to communicate with each other.
h. A centralized network architecture can be used.
i. A decentralized network architecture can be used.
j. A sensor network has a limited geographic range due to the resource-constrained communication technology.
k. In a sensor network, nodes are stationary.

Clarifications: The aim of a sensor network is to transport data to a sink, which then can communicate with a back-end system. If sensors themselves should communicate with a back-end system, they need to have an IP-address. If they only communicate with a sink, they only need a PHY/MAC address. There can be one or more sinks in the network. We have discussed network technologies aimed at sensor networks that have both centralized and decentralized network architectures. A sensor network that uses Argos can cover the whole Earth, so sensor networks do not have to have a limited geographic range. Sensor nodes can be mobile (for example, on animals).

4. Which of the following statements are true for **Delay- and Disruption Tolerant Networks (DTNs)**? (3p)

a. The DTN technology allows multi-hop data dissemination.
b. The DTN technology only allows single-hop data dissemination.
c. The DTN-specific protocols are implemented on the PHY/Link layer in the TCP/IP model.
d. The DTN-specific protocols are implemented on the Network layer in the TCP/IP model.
e. The DTN-specific protocols are implemented on the Transport layer in the TCP/IP model.
f. The DTN-specific protocols are implemented on the Application layer in the TCP/IP model.
g. In a DTN, there must be some stationary nodes.
h. In a DTN, there must be some mobile nodes.
i. DTNs can be used for time-critical applications.

j. DTNs can be used for non-critical applications.
k. Licklider is a DTN-specific protocol.
l. JANUS is a DTN-specific protocol.
m. Gossiping is a DTN-specific protocol.

Clarifications: DTNs are so called overlay networks, where all protocols are implemented on the application layer. DTNs can only be used for non-critical applications since they use a store-and-forward principle. Packets can hop between multiple DTN nodes, and there must be some mobile nodes that can create new links. Licklider is a DTN-specific protocol that has the same purpose as TCP. Gossiping can be used in DTNs, but it is originally developed for sensor networks.

5. Which of the following statements are true for **wireless underwater networks?** (3p)

a. Acoustic communication on low frequency bands can be used for long-range communication.
b. Acoustic communication on high frequency bands can be used for long-range communication.
c. Radio and optical communication links are not suitable for wireless underwater communication.
d. Argos can be used for underwater sensor communication.
e. JANUS can be used for underwater communication.
f. LoRa can be used for underwater communication.
g. Localization of nodes can be performed with GPS.
h. Localization of nodes is not possible underwater.
i. Multi-hop communication can be used in underwater networks.
j. Only single-hop communication can be used in underwater networks.

Clarifications: Acoustic links are good for underwater communication, but also radio and optical wireless links can be used. JANUS is a protocol developed for underwater communication. Argos can not be used underwater, for a node to send data to an Argos satellite, it must be above the sea level.

6. Describe the general network architecture and main protocols for **WiFi 6/7 mesh.** (5p)

Solution:

General network architecture: Wifi 6/7 mesh includes mesh nodes, which all act as access points, and host nodes, which connect to the Internet via the mesh network. A host node can connect to any of the mesh nodes. The mesh nodes forward data between each other until it reaches either the gateway or a host node. One of the mesh nodes must be connected to the Internet, and this node is called the Portal.

Main protocols: There are several protocols in the network, where the main protocols are as follows. The MAC protocol for a host that sends data to a mesh node is contention-free and based on OFDMA. The mesh nodes send data between each other using a deterministic channel access mechanism, called Mesh Coordinated Channel Access (MCCA) which is separate from the host communication. The routing protocol (path selection protocol) for mesh nodes is called Hybrid Wireless Mesh Protocol (HWMP), which provides both a proactive and a reactive path selection mechanism. The proactive mechanism is based on a tree-based approach, and the reactive mechanism is based on AODV. Both mechanisms can co-exist in the same network.

Clarification: You do not need to write all the above to get full points. For full points, you need to describe WiFi 6/7 mesh, not just general things about mesh networks. In your answer, you need

to describe the general network architecture, and an explanation of how data is forwarded between mesh nodes, and how users connect to the network.

7. What is a **Cloud-based RAN (C-RAN)**? Also, describe the general system architecture for a C-RAN. (5p)

Solution:

A C-RAN can be used in 4G/5G cellular networks. In a traditional RAN (called Distributed RAN), the Baseband Unit (BBU) and antenna units, which handles the lower radio layer processing, are placed in the base station. In a C-RAN, the BBU and antenna units are separated. The antenna units, called Remote Radio Heads (RRH), are placed in the base stations. A BBU can handle multiple RRHs and it is placed in a cloud data center (called the central station). Multiple BBUs can be placed in the same data center.

Clarification: For full points, you need to know and describe the above parts.

8. Describe what a **Space Relay** is and when it is used. (5p)

Solution:

Space relays are used in inter-planetary communication. A space relay is a satellite that circles a planet, and that can help to "bend" a communication link (for example, an optical link) to reach the other planet when there is no direct line-of-sight between the sender and the destination. This means that a space relay acts as an intelligent repeater for data communication.

Clarification: Space relays are not specifically developed for DTNs, but a space relay can of course have storage and be used as a DTN node. If you just describe the basics about DTNs, you do not get points. For full points, you need to know what a space relay is used for and that it is a satellite.

Part B: Performance analysis problems (30 points)

9. You are analyzing a client-server application, and you use **a queueing model with one server** for the server part of the application. In your model, new requests arrive according to a Poisson process with mean 8 requests per second. Also, the average service time for each request is 0.1 seconds. Finally, you assume that all requests and service times are independent of each other. (15p)

(a) Your first assumptions are that **the queue is unlimited and that the service times are exponentially distributed**. What is the average utilization of the server, the throughput, and the average response time, *T*, for requests that are served based on these assumptions? (5p)

Solution:

With these assumptions, you model the system as an M/M/1-system with unlimited queue, where λ = 8 requests per second and *E[X]*=0.1 seconds (μ = 10 requests per second).

Utilization = $\bar{N}_S = \lambda \cdot E[X] = 0.8\ (= \rho)$ (Little's theorem)

Throughput = λ = 8 requests per second (there is no blocking in the system).

$T = E[T] = \frac{1}{\mu \cdot (1-\rho)} = 0.5\ seconds$ (formula for M/M/1-systems with unlimited queue).

(b) Your next assumptions are that **the queue is unlimited and that the service times are deterministic**. What is the average utilization of the server, the average throughput, and the average response time, *T*, for requests that are served based on these assumptions? (5p)

Solution:

With these assumptions, you model the system as an M/G/1-system with unlimited queue, where λ = 8 requests per second and *E[X]*=0.1 seconds (μ = 10 requests per second). Also, since the service times are deterministic, *Var(X)* = 0.

Utilization = $\overline{N}_S = \lambda \cdot E[X] = 0.8\ (= \rho)$ (Little's theorem)

Throughput = λ = 8 requests per second (there is no blocking in the system).

$T = \frac{\overline{N}}{\lambda}$ (Little's theorem)

$\overline{N} = \rho + \frac{\rho^2 + \lambda^2 \cdot Var(X)}{2 \cdot (1-\rho)} = 0.8 + \frac{0.8^2 + 8^2 \cdot 0}{2 \cdot (1-0.8)} = 2.4$ (Pollaczek-Kinchine formula)

$T = \frac{2.4}{8} = 0.3\ seconds$

Note: Several of you have established that since *Var(X) = 0* then $E[X]^2 = E[X^2]$, but then you have not used this fact correctly in the Pollaczek-Kinchine formula.

(c) Your final assumptions are that there is **no queue and that the service times are exponentially distributed**. What is the average utilization of the server, the throughput, and the average response time, *T*, for requests that are served based on these assumptions? (5p)

Solution:

With these assumptions, you model the system as an M/M/1/1 system, that is an Erlang system with 1 server (with no queue).

In this case, a request is blocked if the server is busy when the request arrives.

Utilization = $\overline{N}_S = \lambda_{eff} \cdot E[X]$ (Little's theorem)

$A = \rho = \frac{\lambda}{\mu} = 0.8$

$\lambda_{eff} = \lambda \cdot \big(1 - P(the\ server\ is\ busy)\big) = \lambda \cdot (1 - P_B)$

$P_B = \frac{\frac{A^1}{1!}}{\sum_{j=0}^{1} \frac{A^j}{j!}} = \frac{0.8}{1+0.8} \approx 0.44 \Rightarrow \lambda_{eff} = 8 \cdot (1 - 0.44) = 4.44$ requests/second

Average utilization = $\lambda_{eff} \cdot E[X] = 4.44 \cdot 0.1 = 0.44$ (that is, the same as the probability that the server is busy, P_B)

Throughput = $\lambda_{eff} = 4.44$ requests per second.

The average response time, *T*, for requests that are served is *E[X]=0.1 seconds*, since there is no queue (so no waiting time).

Note: Offered and Carried traffic (as in the formula sheet) are not the same as utilization of the server or throughput.

10. You analyze a Multiple-User Multiple-Input, Multiple-Output (MU-MIMO) based WiFi-system where the access point has 3 antennas. Using space-division multiplexing and advanced coding, each antenna can be used by one or more users. There are currently 4 users (A-D) connected to the access point. (15p)

The MAC protocol is based on a basic contention-free and time-slotted system. Each time slot is 100 milliseconds. During a time slot, a user can send up to 1000 kbits (If the user wants to send less than 1000 kbits, this transmission still uses the full time slot). Before each time slot a user notifies the access point if it has data to send. We can assume that this is done on a separate channel with no extra time delays. The access point distributes the antennas to all the users that have data to send, and then the users send their data during the time slot without any data collisions.

Since there are 3 antennas and 4 users, there will be time slots where all users have data to send. In these time slots, two of the users need to share one antenna. These users are randomly selected, and they will get half the capacity of the communication channel during that specific time slot using a contention-free procedure. So, there are never any risks of data collisions or blocking of requests.

Also, during time slots where there are less than three users that want to send data, one or more users are randomly selected to get more than one antenna, so that no antennas are unused during the time slot. However, in this case, no users share an antenna. This means that if there is only one user that wants to send data in the time slot, this user gets all three antennas (with three times the capacity). If there are two users, one of these users gets two antennas, and the other user gets one antenna.
Consider the following scenario: One user (User A) has 10 Mbits data to send. Therefore, User A needs to use several time slots after each other until all data is sent. The other users (Users B-D) are independent of A and each other, and they only want to use single time slots. User B wants to send data in a time slot with probability 0.5. User C wants to send data in a time slot with probability 0.6. User D wants to send data in a time slot with probability 0.8.

In average, how long time will it take for User A to send its data?

Solution:

What do you know?

- 3 antennas, 4 users (A-D)
- 1 antenna during a time slot = 1000 kbits (1 Mbit) data transmitted, 2 antennas = 2000 kbits data, 3 antennas = 3kbits data, 0.5 antenna = 500 kbits data.
- Each time slot is 100 milliseconds long.

- Antennas are randomly selected among users that request an antenna in a time slot.
- User A has 10 Mbits data to send = 10 time slots with 1 antenna. User A requests antennas until all data is transmitted.
- User B, User C, and User D all only request antennas for single time slots.
- P(User B requests an antenna) = 0.5; P(User C requests an antenna) = 0.6; P(User D requests an antenna) = 0.8

What do you want to calculate first?
- Calculate the average number of antennas that User A receives in a time slot

The number of antennas that User A receives in a time slot depends on how many other users that request an antenna in this time slot.

Table of possible outcomes for each time slot (conditioned on that A wants an antenna):

User B	User C	User D	Probability of outcome	Average number of antennas for A
No	No	No	$(1-0.5)\cdot(1-0.6)\cdot(1-0.8)=\mathbf{0.04}$	3
Yes	No	No	$(0.5)\cdot(1-0.6)\cdot(1-0.8)=\mathbf{0.04}$	$0.5\cdot 1+0.5\cdot 2=1.5$
No	Yes	No	$(1-0.5)\cdot(0.6)\cdot(1-0.8)=\mathbf{0.06}$	$0.5\cdot 1+0.5\cdot 2=1.5$
No	No	Yes	$(1-0.5)\cdot(1-0.6)\cdot(0.8)=\mathbf{0.16}$	$0.5\cdot 1+0.5\cdot 2=1.5$
Yes	Yes	No	$(0.5)\cdot(0.6)\cdot(1-0.8)=\mathbf{0.06}$	1
Yes	No	Yes	$(0.5)\cdot(1-0.6)\cdot(0.8)=\mathbf{0.16}$	1
No	Yes	Yes	$(1-0.5)\cdot(0.6)\cdot(0.8)=\mathbf{0.24}$	1
Yes	Yes	Yes	$(0.5)\cdot(0.6)\cdot(0.8)=\mathbf{0.24}$	$0.5\cdot 1+0.5\cdot 0.5=0.75$

(A good way to check that you have done the correct calculations is that the sum of all outcome probabilities must be 1.)
The average number of antennas for User A in a time slot (using law of total probability):
$0.04\cdot 3+0.04\cdot 1.5+0.06\cdot 1.5+0.16\cdot 1.5+0.06\cdot 1+0.16\cdot 1+0.24\cdot 1+0.24\cdot 0.75=1.15$

This means that User A in average can send $1.15\cdot 1000\ kbits=1150\ kbits$ data in each time slot.

Therefore, the average number of time slots that User A needs to send 10 Mbits of data:
$\frac{10\ 000\ kbits}{1150\ kbits}\approx 8.7\ \Rightarrow 9$ time slots must be used.

So, **the average total time to send 10 Mbits of data is 900 milliseconds**.

Clarification: To get full points, you need to do all the steps correctly. Some students have used the number of antennas that A gets in a time slot (0.5, 1, 2, 3) as "outcomes", and that is perfectly fine. In this case, the calculations look slightly different, but the result is the same. The number of reduction points depends on where your logic goes wrong. Students have many different types of non-correct solutions. If you have a good starting approach, but then do the wrong type of calculations, you get about 5 points. You do not get any points for just listing the things you know. Several students have missed that full time slots are always used independent on how much data that is sent in the time slot.

Part C: System design scenarios (30 points)

For each of the scenarios given below, there are some general recommendations for the system design in this scenario, and some solution examples, which were given full points on the exam. The solutions have been only minor edited, fixing typos and small errors that were not given point reductions.

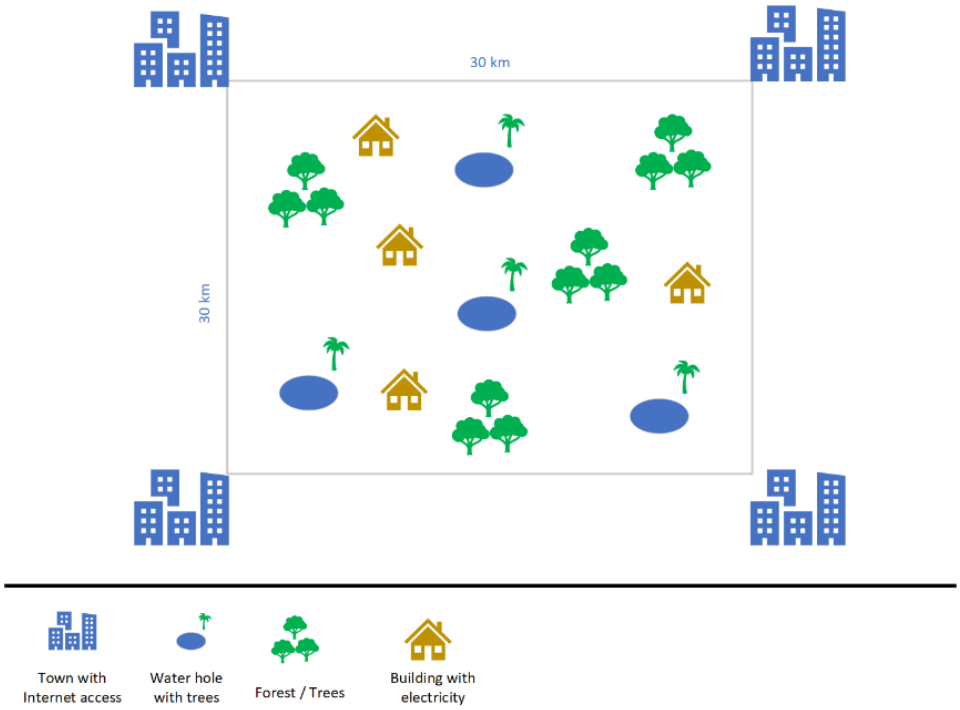
Grading: You get full points if your system solution is realistic and includes all important parts, and if your arguments are correct and coherent. You get 8-14 points if you have provided a coherent system design with correct networks technologies, but your solution lacks some details (the reductions depend on what you lack in your solution). You get 0-7 points if your chosen network technologies are not suitable, your solution lacks large parts of the important challenges, or your solution is general and thereby not including technical details for this specific scenario.

11. The national parks in Africa experience lots of problems with illegal hunting. You have been given the task to help a national park in South Africa with a surveillance monitoring system. A simple map of the park is illustrated in the figure below. The area not marked in the map is a savanna, made up of large expanses of grasses interrupted by scattered shrubs and small trees. There are several different types of animals in the park, including lion, leopard, rhinoceros, elephant, and African buffalo (The "Big 5" for hunting), in the park, and they all visit the water holes at regular times. At other times, they are spread out over the park.

Your task is to design the network architecture for a surveillance system for the park. The surveillance system should detect potential hunters. If a hunter is detected, the position of the hunter should be estimated, and a warning signal should be sent to the headquarters (placed in the top right town in the figure).

You have access to all types of sensors, including heat and motion sensors, and various small microprocessor computers (Raspberry PIs). Also, you have access to trail cameras (small cameras that take low-quality images when actuated). All equipment can use all types of communication technology standards. There is no cellular coverage within the national park (only in the towns). Starlink is available in South Africa.

You have a decent budget for your solution, but it is preferrable if the system can have low maintenance and be deployed with minimum interference with the wildlife. Also, the park appreciates low operational costs.



The scenario: This is a highly realistic scenario, and similar systems have already been deployed, or are investigated[1].

Recommendations: The application must be non-critical, since it is resource-constrained, and the scenario is a typical LoRa scenario. Therefore, the main challenge in this system design problem is to get all the details correct: Where to place sensors/cameras, how to power the sensors/cameras, where to place LoRa gateways, how to power LoRa gateways, the backhaul link from gateways to network server, how to detect illegal hunters with your system etc. Also, you need to describe a solution for the localization of potential illegal hunters (GPS or other solution). If you propose to use any type of edge computing (for example, with a Raspberry Pi) connected with a trail camera, you need to describe how this Raspberry Pi will be powered, since a battery may not be enough unless you plan to change batteries often.

Some of you have suggested that the surveillance system only monitors the roads leading into the park. This is a very limited solution, and illegal hunters will always find ways to get around this (which is happening today).

Note that in standard LoRa, the gateways do not have a mesh structure, each LoRa gateway must have a backhaul link to the Internet (where the network server is). One of the project groups found LoRaWAN Mesh, but you need to specifically say that you use this technology to get full points. Also, you need to provide ONE complete and correct network system solution, not only describe different alternatives, to get full points.

You can use Starlink as backhaul but note that Starlink is mains powered. There are different home-built solutions with solar panels and larger batteries (like car batteries), but Starlink is not developed as a solar-powered backhaul. The same is valid for different types of radio links, like microwave links. There are buildings with electricity in the park, which can be used for the gateways. Argos is not

[1] https://www.cambridge-design.com/case-study/zsl-instant-detect-2-0/

suitable as backhaul link, since Argos only allows for an extremely low bit rate and periodic data transmission.

Solution example 1:

Requirements: Detect potential hunters; Alert the headquarters fast enough for them to act on it while the potential hunters are still there; The alert should include the location of the hunters; Good if the headquarters can identify if it's a potential hunter or animal in the alert to avoid unnecessary controls; Resource-constrained network.

Challenges: The backhaul since there is no cellular coverage in the park; Power for equipment.

Solution: The scenario is a monitoring scenario, but with some requirements on the latency since the headquarters needs the location of potential hunters fast to act on it. However, it isn't a time-critical scenario since the latency doesn't have to be super low (as it needs to be when for instance controlling cars on roads). Therefore, the scenario can be classified as a not-so-time-critical scenario with low bit rates, low-high latency and low power consumption. The low power consumption part is important since it's a resource-constrained network.

We will use LoRa in this solution since it's a technology suitable for not-so-time-critical applications and has a long radio range. The LoRa network server will be placed in the headquarters with access to cellular coverage and possibly WiFi since it's reasonable for headquarters to have that. The network server can also be mains powered there since it's in the city. LoRa base stations will be placed near the yellow buildings that have electricity and one base station will be placed in the top right corner and one in the lower right corner to ensure coverage over the whole park. These stations will be mains powered since both the yellow building and the stations on the city borders will have access to electricity. For the backhaul I suggest using microwave links to the city where there is cellular coverage.

In the park there will be LoRa class A sensors placed in suitable spots to monitor motion and heat. The sensors will be connected to a small microprocessor and a small low-quality camera. When the sensors detect motion and/or heat, the camera will take a picture. The microprocessor will also estimate how far away the motion/heat is from the sensor. This estimation will be based on the information from the sensor and a GPS-system to calculate approximate coordinates. The alert will then be sent, and this will include a low-quality picture and coordinates. LoRa only works with low bit rates, but since the picture is of low quality it will work to send it over LoRa. When the alert comes to the headquarters, they can hopefully identify from the picture if it's a potential hunter or an animal to avoid going out on an alert for just an animal. This will make the operational costs smaller. The alerts will be stored in a private cloud which the headquarters can access from their computers/phones.

The sensors will be LoRa class A sensors as said, and this means that they will use modified ALOHA for its communication. The device will broadcast the alert at any time and then enter sleep mode and wake up after 1 and 2 seconds to listen for an ACK. This means that multiple base stations can get the alert and then the base stations decide which station to send the ACK. This makes the probability of the alert reaching a base station higher and if the device doesn't get an ACK, it will send again. Sleep mode is also crucial since it's resource-constrained and LoRa also uses half-duplex communications which also saves power.

The little system of sensor, microprocessor and camera will use energy harvesting with a rechargeable battery. Solar panels will be used, and this is ideal for the climate on the savanna where

the sun shines every day (almost). At night the sensors will use power from the battery that gets charged during the day. This will result in low maintenance and low operational costs.

This system has low maintenance, low operational costs and I would it also say can be deployed with little interference to the wildlife since the base stations are located where there already is civilization and the sensor systems are small and preferably placed up in trees which there is over the whole savanna.

Solution example 2:

This is a non-critical scenario since there can be short delays without the system breaking down, i.e. when an illegal hunter is detected most of the time will anyway be put on moving to that location and finding the illegal hunter. The environment poses a couple major challenges, some of which are: lack of cellular connectivity, large areas to cover and making the sensors non-invasive and hard to find for the hunters.

I believe that a Low Power Wide Area Network (LP-WAN) would be fitting for this scenario, more specifically Long Range Wide Area Network (LoRaWAN or LoRa), since the alternatives for non-critical, large geographic footprint sensor networks either have high running costs (e.g. Argos, which also provides very low data rates) or have requirements which are unavailable (NB-IoT which requires cellular connectivity). I would use multiple gateways because one wouldn't be enough to cover the whole area since their max range is around 10 km (depending on conditions). The gateways need a more robust source of energy, and therefore I would put one on the roof of each of the houses, which I assume are mains powered. I would then put the network station on the outskirts of the city where the headquarters is (top right city), which would then use the cellular network in the city to send the data to the headquarters. The gateways would then use microwave links, which have sufficient range to cover the whole park, to connect to the network station. Since microwave links need line-of-sight, the gateways would have tall poles from the roofs of the buildings.

The LoRa Class A devices would be coupled together with the heat and motion sensors as well as the trail cameras, this to be able to transmit the information to the gateways. Since most of the mentioned animals probably only drink during the day I would put the trail cameras by the watering holes, which I assume many illegal hunters go to. I would then put the heat and motion sensors by common sleeping spots too since the cameras can't see during night. The sensors, Raspberry PIs and trail cameras are all resource constrained. This means that they would be battery powered. To save battery, the LoRa Class A devices would sleep until the sensor/camera gets triggered. Also, the built-in sleep protocol that is used during transmissions would also be used; when the Class A device sends data to its closest gateways, two receive windows (for the ACK) are opened and it sleeps between them. The gateways send the message forward to the network server which decides which of the gateways will send the ACK.

I would put small solar panels on the devices to prolong their battery life, which is fitting thanks to the being close to the equator, providing the solar panels with rather consistent year-round sun. For the sensors put on trees I would draw a cable to the tree-top and put the solar panel there to get more sun. This is one part that would decrease the operation costs and time needed for switching batteries.

To decrease the amount of data sent, I would put the Raspberry PIs on the cameras in order for the connected LoRa device to only send data when the photo is of a human (the Raspberry PIs uses image recognition). This would be a kind of Edge Computing.

Their MAC-address of the sensors can be hard coded to be mapped to the coordinates of their stationary locations. This would mean that, upon receiving data from a specific MAC-address, the coordinates are indirectly provided.

Solution example 3 (only lacking a power solution for the edge computing):

Challenges: Wide area; Moving objects of surveillance; No connectivity; Scarce options for electricity; Minimum interference with wildlife

QoS: Suitable for sensors; Not-so-time critical; Low maintenance; Low cost

QoE: System identifies hunters correctly; System covers area efficiently; System alert HQ upon identifying hunters; System alerts with sufficient timing so that HQ can dispatch personnel to hinder hunters efficiently

Proposed solution: As the goal of the system will be to cover a wide area and warn for hunters, while minimizing interference with wildlife and catering to low maintenance and low budget needs, my proposition is to implement a LoRa wireless sensor network.

LoRa stands for Long Range and is based off of a LP-WAN and Long Range WAN, that is thus low power and long range, suitable for resource-constrained scenarios. To implement LoRa we need access points. The access points range depends on infrastructure and the landscape, but in rural areas approximately 10-15km. I will thus suggest placing access points by the three houses closest to the border of the park, which should cover the entirety of the park (as it is 30x30km). Hopefully, these buildings will be willing to cooperate with placement. To be able to operate these access points, a network server will have to be installed in which I would place it by the fourth building in the middle. The network servers' goal is to coordinate the access points and assign which access point will ACK a package that has arrived from a sensor. Communication between the network server and the three access points can be done through an antenna and point-to-point links. Power supply for the four stations can hopefully be implemented through the houses' energy supply.

To connect the LoRa wireless sensor network to HQ, a backhaul will need to be implemented. As I know Starlink is available in South Africa, I would suggest placing a Starlink antenna by the house with the network server. As LoRa data rates are quite low, Starlinks uplink capacity should be sufficient. This will then be able to connect to HQ through the core network where the alerts can be shown through the user interface of a web-application. To be able to localize which sensor made the alert, I suggest using the devices manufacturer's ID, and have a back-end system wherein the ID's and the sensor's locations are stored and easily retrieved.

It is important to note that LoRa is not a network architecture for time-critical applications as it is resource-constrained to lower power usage and maintenance. Even though hunting can feel like a time-critical scenario, in the context of networking it is not, as an alert that comes a few seconds or tens of seconds later instead of milliseconds or seconds will not greatly affect HQs capacity to dispatch personnel.

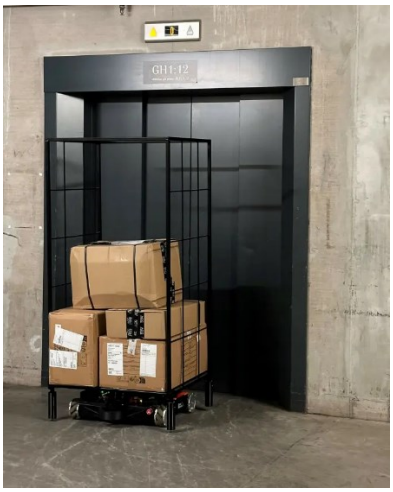
As goes for the sensor network itself, class A LoRa sensors can be placed within the reach of each access point, which means the whole park. I would suggest placing sensors by water holes, trails and other points of interest if there are (i.e. common points of gathering or common places where illegal hunters enter the park). The sensors will use a thermal and motion camera, alongside an audio recorder. As LoRa sensors have limited data rates and recourses, I would suggest placing a Raspberry PI by each sensor to aid these tasks but also to perform edge computing. The Raspberry PI will analyze the data and alert the sensor when it identifies humans (i.e. human shapes, human noise or

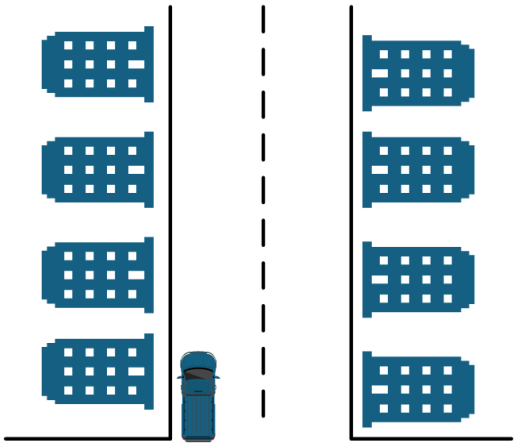
gunshot noise). The evidence will then be compressed and sent through the LoRa sensor to the AP. The MAC-protocol for Class A LoRa devices is like ALOHA and can send data when it wants and sleeps the rest of the time. It will however wake up 1 and 2s after the data has been sent to be able to receive an ACK from an AP. The power supply of the LoRa sensors will be battery since energy harvesting may disturb the surrounding nature.

12. Your task is to help Scania with their new project that investigates mobile robots for "last meters" deliveries in cities. The image below to the left shows Scania's new mobile robot prototype Carrie. Carrie can deliver packages and boxes from a truck (parked on the street) to the door of a building on this street (the image shows an elevator, but your project targets outdoor deliveries). With Carrie, trucks do not have to drive in all the narrow streets of a city. Instead, trucks can be parked at the beginning of a street, and then Carrie delivers the packages the last meters. With Carrie, city logistics and deliveries become more sustainable, both from a transport perspective and a human perspective, since the trucks can be parked further away from the destination building and drivers to not have to deliver heavy packages the last distance.

Currently, Carrie is remote-controlled by the driver with a direct radio communication link (Line-of-sight). However, Scania wants to investigate how Carrie can be self-driving using a driving control AI system in the cloud. Carrie will have various on-board cameras and sensors that can be used by the AI system for giving driving decisions.

Your task is to design a network system for Carrie's new self-driving mechanism. The targeted scenario is illustrated in the image below to the right. The truck parks at the beginning of a street in the city (we can assume that it is a Swedish city). The targeted streets are up to 100 meters long. Carrie should be able to safely deliver packages to all front doors on this street by driving on the sidewalks (we assume that the sidewalks are broad enough for Carrie). Carrie will have a fail-safe mechanism for short losses of connectivity, so that accidents do not occur if there is a short period of bad connectivity. For longer streets with multiple deliveries, the truck can have multiple Carrie robots to decrease delivery times. If so, the robots will deliver packages at the same time.





Scenario: Carrie and other types of transportation robots are currently developed or in operation, but probably mainly for indoor scenarios. So, this specific outdoor scenario is somewhat fictional but interesting from a network system perspective.

Recommendations: This is a typical time-critical application. The scenario is set outdoors, but Carrie is probably rather slow moving, so this is not a vehicular network scenario, it is more a cloud robotics scenario.

For full points you need to give a suitable complete network system solution, with correct logic and reasoning. Several of you have mixed networks technologies that do not work together. Also, for full points, you should give a solution for the positioning that is required (depending on the chosen network technology).

Since the longest distance between a truck and a Carrie is 100 meters, with line-of-sight as illustrated in the figure, it would be feasible to suggest a WiFi 6/7 network with the access point in the truck (or on the roof of the truck). However, no students with full points suggested this solution.

The scenario states that the AI is placed in the cloud. Your solution must allow for Carries to access the cloud with low latencies (due to the time-critical scenario). If you suggest any type of edge/fog computing solution, this solution must work with the chosen network technology, and you need to specify how it will work.

If you say that you will use a VANET, then you need to describe how this VANET will work. You cannot just say that your solution is a VANET based on 5G and then leave it there.

If you suggest a private 5G network, it should be an integrated private network, using network slicing from a national mobile operator. It is not feasible that Scania should build their own private 5G network and get their own frequency license. Several students mixed up what a private 5G network is, and how it is built.

Solution example 1:

This is a time-critical scenario as the AI system responsible for Carrie's autonomy is deployed in a cloud environment and isn't run locally. This means that the scenario requires reliable, high bitrate, low latency communication between the device and the datacenter.

For each delivery route there are three main actors in the network: The truck, the Carrie (or multiple Carries) that belongs to the truck and the datacenter.

The communication between Carrie(s) and the truck is simple, as the information that needs to be exchanged is the locations of both parts (assuming the truck will want to drop different Carries in different streets in the future and move between streets as they deliver the packages) as well as the delivery status from Carries. The locations as well as the delivery status would be sent to and stored in a datacenter from which both parties can fetch the necessary information.

The communication between a Carrie and the datacenter on the other hand is not as simple. Although the information necessary for the AI algorithm to make decisions and enable autonomous driving for Carrie isn't as complex as it is for a car, it is still quite complex involving images and/or LIDAR data as well location updates from Carrie. It's mentioned that there are fail-safe mechanisms, however this is still a real-time application where the reliability of the communication channel is very important and the requirements on latency and capacity is strict.

A suitable solution for this kind of application is utilizing an integrated (with network slicing) private 5G network. This kind of network architecture provides the Carrie(s) with dedicated capacity necessary for the real-time communication between the actors in the network. 5G also supports URLLC which is important in this kind of scenario. The 5G network, thanks to its MAC protocols, can easily allow simultaneous access to network capacity to several Carries working in parallel. In this

network, each Carrie would be identified with an eSIM and would also be equipped with a GPS module for tracking the location. As the application is meant for last mile delivery in cities, the coverage is not a problem. The truck itself doesn't exactly need internet access as a mobile application on the driver's phone is more than sufficient for communication between the Carrie(s) and the driver.

The placement of the datacenters also deserves significant consideration. In this kind of application, fog computing can be very helpful to decrease the latency of the communication between datacenters and the Carries. Communicating with a huge central datacenter in California is not the best idea if packages are being delivered in Stockholm city. It's ideal that the computations (in this scenario the inference from the AI system) happen as close to the Carrie as possible, which in this scenario could mean in the same city. Where the delivery statuses are sent to and stored in are not as critical as they are not involved in complicated computations.

Regarding, the Carrie(s) would run on battery during the day and needs to be changed during the night. The other actors in the network (driver's phone and the datacenter(s)) already have their own power solutions.

For the Carries to safely and correctly deliver packets, they need to have some sort of built-in localization mechanism (in addition to the cameras and sensors that allow the AI to steer them more precisely). This can be achieved through incorporating GPS sensors in the Carries. The Carries must also have a unique physical address, to send and receive data to/from the AI in the cloud. Location and ID can then be added to transmitted data packets from a Carrie, so that the AI can keep track of different Carries, is always aware of their positions and can give informed instructions about driving decisions. The general speed of the Carries should be relatively slow, to ensure that moving obstacles (like cars, people and animals) are detected in time for collision-avoidance algorithms to be set into motion.

There are, of course, many other solutions that could be deployed in this scenario. In my opinion, however, this is a good solution that meets the requirements of Scania, without unnecessarily high costs.

Solution example 2:

The system for this product will need a time-critical network. This is because it needs to send a huge amount of sensor data in real time and constantly receive instructions from the cloud server. Since the solution only will be used in Swedish cities, we have the advantage of being able to use 5G, which has providers covering almost all of Sweden. 5G technology can provide very high bandwidth in a duplex manor and won't require us to develop our own network infrastructure. Because we don't need to develop our own network also makes this solution very easily scalable. The only downside in this case would be the need for a subscription from a network provider.

Each unit of the Carrie robot will be equipped with a 5G networking device and corresponding sim-card for identifying itself on the network. The robot should already have energy for powering the device and a central computer that can gather the sensor data and send it. To minimize latency, the cloud server should optimally be placed somewhere in central Sweden and use the same mobile net operator as the Carrie units.

For the driver to send instructions on where the packages should go, they will use a mobile application developed for this specific purpose. When a Carrie is loaded with its packets and turned on the driver will select its id from a list of their Carries in the application and then input the address.

A satellite GPS map will show up with the location and the driver selects where exactly the destination is. These instructions are then sent to the same cloud server, which plans a path for Carrie. When it is ready the driver presses start, and the server starts sending commands based on the sensor data.

Solution example 3:

Let's break down the solution,

Requirements of the system: Robot Carrie need a communication system for self-driving; Safely deliver packages to all front doors on this street by driving on the sidewalks; Driving control AI system at the cloud; Fail-safe mechanism for short losses of connectivity; Not using: direct radio communication link (Line-of-sight); Location: Swedish city; Multiple Carrie robots to decrease delivery times

Challenges: Real-Time communication; Carrie need to avoid running into people and objects on the street; 100 meters of communication link - short distance communication; Redundancy communication link; Security; Deliver packets at the same time

*QoS***:** Low latency; Quick reactions for Carrie to avoid collision of objects; Packages need to be safe from theft along the way

*QoE***:** People need to be able to take note that there is a robot on the lose; Notification to customer that a Carrie is at the door ready for delivery.

Since the solution needs to be cloud based, be able to perform in a time-critical scenario, have good latency for short distance (100m) and provide a good communication from an autonomous vehicle (Carrie) to an infrastructure (the cloud) a cellular solution can be applied with an C-V2X approach. Since most of all the cities in Sweden are today provided with 5G this will be a good network system to be used between Carrie and the cloud. Backhaul that can be used is wired optical communication. Since the system demands redundancy as a safety aspect, 4G-LTE is a good fallback solution, both 4G/5G has good coverage in most of the cites in Sweden according to teleoperators such as Telia. Fog computing will be a good thing to consider to provide in the city, so that the cloud will be closer to the area of need.

A cloud server, provided with AI algorithms, good computing powers need to be performed and therefore good hardware is needed, which will be expensive. Each Carrie-robot need to have an OBU (Onboard-Unit), based out of C-V2X communication protocols, that can transmit GPS-signal of its current location to the Cloud and have a receiver unit that will be able to take commands straight away from the cloud, that send back commands (STOP, FORWARD, LEFT, RIGHT etc). Meaning the OBU needs to have a full-duplex communication link, between Carrie and the cloud, so that messages can be sent both ways, at the same time. Carrie notifies the cloud of current scenario, and the cloud notifies Carrie on how it should act. Also, once Carrie reaches the door, it will stop, through an application the buyer can enter "Yes Carrie is in front of me with my packet", once the cloud receives this information, Carrie will then let the buyer access its cargo and notify the cloud once the cargo has left its carriage so the cloud can give the command of Carrie to return to the truck.

Since the robot needs to be able to detect what location it needs to go to and detect obstacles along the way each Carrie needs to be provided with good IoT-sensors to avoid collision at the street. Since the Carrie robot needs to perform work during a maximum 200m distance, multiple times, if the same robot needs to make several deliveries at the same street the power supply of the robot needs

to have its own battery. This battery needs to be able to be rechargeable. Each truck will have its own power-recharging station for the Carrie robots. Once the truck drive, Carries are recharged.